

# Divisor-Sensitive Bounds for Forbidden Subset Sums

## Abstract

Let  $F(E, \tau)$  be the largest cardinality of a set  $A \subseteq \{1, \dots, E\}$  having no subset with sum  $\tau$ . We prove a divisor-sensitive upper bound for  $F(E, \tau)$ , with an explicit error governed by the largest three-term-progression-free subset of an interval. The bound is asymptotically sharp on infinitely many families of linear targets. As an application, if  $f_c(n) = F(\lfloor cn \rfloor, n)$ , then  $f_c(n)/n$  fails to converge for every  $0 < c < 1$ , answering a question of Erdős and Graham. At  $c = 1/2$  we obtain the two exact subsequential limits  $1/4$  and  $1/6$ .

For a finite set  $A$  of positive integers, let

$$\Sigma(A) = \left\{ \sum_{a \in B} a : B \subseteq A \right\}, \quad F(E, \tau) = \max\{|A| : A \subseteq [1, E], \tau \notin \Sigma(A)\}.$$

Write  $r_3(N)$  for the largest cardinality of a subset of  $[1, N]$  containing no nonconstant three-term arithmetic progression, and put  $L_P = \text{lcm}(1, \dots, P)$ .

Our starting point is the following finite form of Alon's bounded zero-sum theorem [1, Proposition 2.5]. If  $k \geq 2$  and  $A \subseteq \mathbb{Z}/N\mathbb{Z}$  satisfies

$$|A| \geq \frac{N}{k} + (1 + \sqrt{3(k-2)})r_3(N),$$

then some nonempty  $B \subseteq A$  has  $|B| \leq k$  and  $\sum_{b \in B} b = 0$  in  $\mathbb{Z}/N\mathbb{Z}$ .

**Theorem 1.** *Let  $P, d, E, \tau$  be positive integers such that*

$$L_P \mid d, \quad d \mid \tau, \quad dE \leq \tau.$$

*Set*

$$q = \frac{\tau}{d}, \quad K = \left\lfloor \frac{(P+1)q}{E} \right\rfloor, \quad T = 1 + \sum_{w=1}^P \left( \frac{d}{w} - 1 \right).$$

*Then*

$$F(E, \tau) < \frac{q}{K} + (1 + \sqrt{3(K-2)})r_3(q) + (T-1)K. \quad (1)$$

*Proof.* Suppose that  $A \subseteq [1, E]$  has cardinality at least the right-hand side of (1). Since  $q \geq E$ , reduction modulo  $q$  is injective on  $A$ . Alon's theorem yields a nonempty set  $B_1 \subseteq A$  with

$$|B_1| \leq K, \quad \sum_{b \in B_1} b \equiv 0 \pmod{q}.$$

The ordinary sum of its elements is positive and strictly smaller than  $(P+1)q$ . Indeed, it is at most  $KE \leq (P+1)q$ ; equality would require  $K$  distinct elements all equal to  $E$ . Thus

$$\sum_{b \in B_1} b = w_1 q \quad (1 \leq w_1 \leq P). \quad (2)$$

Delete  $B_1$  and repeat. Before the  $i$ th application, at most  $(i-1)K \leq (T-1)K$  elements have been deleted, so Alon's hypothesis still holds. We obtain  $T$  pairwise disjoint sets  $B_i$  satisfying

$$\sum_{b \in B_i} b = w_i q, \quad 1 \leq w_i \leq P. \quad (3)$$

Since every  $w \leq P$  divides  $d$ , the definition of  $T$  shows that some weight  $w$  occurs at least  $d/w$  times: otherwise the number of weights would be at most  $\sum_{w=1}^P (d/w - 1) = T - 1$ . The union of  $d/w$  corresponding sets  $B_i$  has sum  $dq = \tau$ .  $\square$

Roth's theorem gives  $r_3(N) = o(N)$  [3]. The finite estimate therefore has the following uniform consequence.

**Corollary 2.** *Fix  $\rho \geq 1$ . Uniformly for  $E \leq \tau \leq \rho E$ ,*

$$F(E, \tau) \leq \min_{d, P \geq 1, \substack{d|\tau, \\ L_P | d}} \frac{\tau/d}{\lfloor (P+1)\tau/(dE) \rfloor} + o_\rho(E). \quad (4)$$

*Proof.* For every pair occurring in the minimum,  $d \leq \rho$  and  $L_P \leq \rho$ . There are consequently only finitely many such pairs. Moreover  $E \leq q = \tau/d \leq \rho E$ , so  $r_3(q) = o_\rho(E)$ . Apply Theorem 1 and take the minimum.  $\square$

For  $y = \tau/(dE) \geq 1$ , the main term in (4) satisfies

$$\frac{yE}{\lfloor (P+1)y \rfloor} < \frac{P+2}{(P+1)^2} E. \quad (5)$$

Thus increasing the divisibility available in  $\tau$  produces a hierarchy of density bounds. In particular, if  $s$  is the smallest positive integer not dividing  $\tau$  and  $L_{s-1}E \leq \tau \leq \rho E$ , then

$$\left\lfloor \frac{E}{s} \right\rfloor \leq F(E, \tau) \leq \frac{\tau/L_{s-1}}{\lfloor s\tau/(L_{s-1}E) \rfloor} + o_\rho(E). \quad (6)$$

The lower bound is given by the multiples of  $s$ .

The bound is sharp on a large collection of rational rays. This is most naturally expressed for the original one-parameter problem

$$f_c(n) = F(\lfloor cn \rfloor, n).$$

**Corollary 3.** *Let  $s$  be a prime power, and let  $d, m$  be positive integers satisfying*

$$L_{s-1} \mid d, \quad s \nmid d, \quad m \geq s.$$

*If  $c = s/(dm)$ , then*

$$\frac{f_c(n)}{n} \longrightarrow \frac{1}{dm} \quad \text{as } n \rightarrow \infty \text{ through } d \mid n, \ s \nmid n. \quad (7)$$

*Proof.* Put  $E = \lfloor cn \rfloor$  and apply Corollary 2 with  $P = s-1$ . For all sufficiently large  $n$ ,

$$\left\lfloor \frac{sn}{dE} \right\rfloor = m,$$

so (4) gives  $f_c(n) \leq n/(dm) + o(n)$ . Conversely, the multiples of  $s$  in  $[1, E]$  have no subset sum equal to  $n$ , since  $s \nmid n$ . Their cardinality is  $E/s = n/(dm) + O(1)$ .  $\square$

For example, if  $c = 2/m$  with an integer  $m \geq 3$ , then the case  $(s, d) = (2, 1)$  and the same upper bound applied without a congruence restriction give

$$\limsup_{n \rightarrow \infty} \frac{f_c(n)}{n} = \frac{1}{m}, \quad \frac{f_c(n)}{n} \longrightarrow \frac{1}{m} \quad (n \text{ odd}). \quad (8)$$

We now settle the question of Erdős and Graham [2, p. 59].

**Theorem 4.** For every  $0 < c < 1$ , the sequence  $f_c(n)/n$  does not converge. More precisely,

$$\liminf_{\substack{n \rightarrow \infty \\ n \text{ odd}}} \frac{f_c(n)}{n} \geq \frac{c}{2},$$

whereas

$$\limsup_{\substack{n \rightarrow \infty \\ n \text{ even}}} \frac{f_c(n)}{n} \leq \begin{cases} \frac{1}{2\lfloor 3/(2c) \rfloor}, & 0 < c \leq \frac{1}{2}, \\ c - \frac{1}{2} + \frac{1}{2\lfloor 3/(2(1-c)) \rfloor}, & \frac{1}{2} < c < 1. \end{cases} \quad (9)$$

Both quantities on the right of (9) are strictly smaller than  $c/2$ .

*Proof.* For odd  $n$ , take the even integers in  $[1, \lfloor cn \rfloor]$ .

Let  $n$  now be even and put  $M = \lfloor cn \rfloor$ . If  $c \leq 1/2$ , apply Corollary 2 with  $E = M$ ,  $\tau = n$ , and  $(d, P) = (2, 2)$ . This gives the first bound in (9).

Suppose  $c > 1/2$  and put  $D = n - M$ . Apply the same estimate to  $A \cap [1, D - 1]$ , with  $E = D - 1$  and  $\tau = n$ . On  $[D, M]$ , the involution  $x \mapsto n - x$  permits at most one element from each two-element orbit, together with its fixed point  $n/2$ . Hence

$$|A| \leq \frac{n}{2\lfloor 3n/(2(D-1)) \rfloor} + M - \frac{n}{2} + o(n),$$

which gives the second bound.

Finally,  $\lfloor 3r/2 \rfloor > r$  for every  $r \geq 2$ . Taking  $r = 1/c$  and  $r = 1/(1-c)$ , respectively, proves the strict inequalities.  $\square$

At  $c = 1/2$  the arithmetic fluctuation can be evaluated on both sides: Corollary 3, first with  $(s, d, m) = (2, 1, 4)$  and then with  $(s, d, m) = (3, 2, 3)$ , gives

$$\frac{f_{1/2}(n)}{n} \rightarrow \frac{1}{4} \quad (n \text{ odd}), \quad \frac{f_{1/2}(n)}{n} \rightarrow \frac{1}{6} \quad (2 \mid n, 3 \nmid n). \quad (10)$$

For completeness, when  $c \geq 1$  complementary pairs give the exact formula

$$f_c(n) = \lfloor cn \rfloor - \left\lceil \frac{n}{2} \right\rceil.$$

Thus the irregularity question is determined for every  $c > 0$ .

## References

- [1] N. Alon, Subset sums, *J. Number Theory* **27** (1987), 196–205.
- [2] P. Erdős and R. L. Graham, *Old and New Problems and Results in Combinatorial Number Theory*, Monographies de L'Enseignement Mathématique, no. 28, Geneva, 1980.
- [3] K. F. Roth, On certain sets of integers, *J. London Math. Soc.* **28** (1953), 104–109.